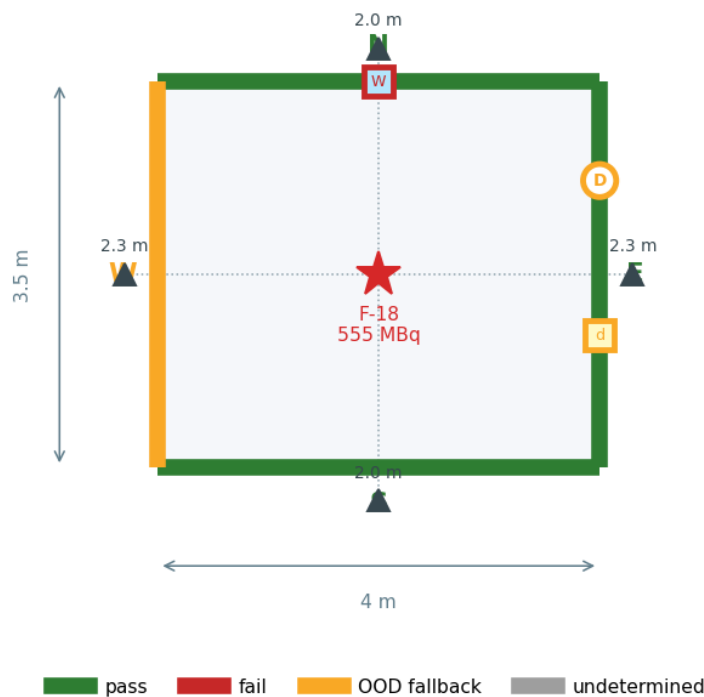


ShieldLab — Room Shielding Report

Check (declared shielding) | generated 2026-07-05 23:24

Inputs

Isotope:	F-18
Activity per patient (MBq):	555
Patients / week:	30
Residence (min/patient):	60
Room (W×L×H, m):	4 × 3.5 × 3
Source position (x,y m):	(2, 1.75)
Framework:	NCRP



Per-barrier results

Barrier	Mat.	Sug. mm	Analyt. B	Surrogate B (95% CI)	Dose	Verdict	Margin
Wall N	concrete	—	0.147	0.0769 [4.7e-03, 1.4e-01]	0.047	PASS	2.13
Wall N · window	concrete	—	0.742	0.574 [3.9e-03, 8.4e-01]	0.351	FAIL	0.285
Wall E	concrete	—	0.323	0.233 [5.6e-03, 3.0e-01]	0.113	PASS	1.42
Wall E · door	concrete	—	0.742	0.742	0.34	FAIL	0.47
Wall E · duct	concrete	—	—	—	—	—	—
Wall S	concrete	—	0.323	0.233 [5.6e-03, 3.0e-01]	0.142	PASS	1.41
Wall W	concrete	—	1	1	0.485	PASS	1.65

Decision-support output — requires review and sign-off by a qualified Radiation Safety Officer / medical physicist. Analytical tier: NCRP-151 / AAPM TG-108 broad-beam model (ShieldLab, validated). Surrogate tier (where shown): Monte-Carlo-trained Extra-Trees with conformal 95% intervals and an out-of-domain guard (thesis, 2026).

Notes: REPRESENTATIVE PET uptake room (TG-108-informed workload). Placeholder for the thesis RSO case study - replace with the real facility data when collected.